DECENCY AND REASONABLENESS PULL BACK ALONG ETALE MORPHISMS

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Editorial note. The source compresses both cases to 'Hence the result'; the injections and the subset bound are carried as explicit obligations.

Allowed standard prerequisites (editorial).

- Etale morphisms are open. Remark remark-recall (same chapter, structural). Used in the fibre comparisons.

Lemma 0.1. *Let S be a scheme. Let $X \to Y$ be an étale morphism of algebraic spaces over S. If Y is decent, resp. reasonable, then so is X.*

Proof. Let U be an affine scheme and $U \to X$ an étale morphism. Set $R = U \times_X U$ and $R' = U \times_Y U$. Note that $R \to R'$ is a monomorphism.

Let $x \in |X|$. To show that X is decent, we have to show that the fibres of $|U| \to |X|$ and $|R| \to |X|$ over x are finite. But if Y is decent, then the fibres of $|U| \to |Y|$ and $|R'| \to |Y|$ are finite. Hence the result for "decent".

To show that X is reasonable, we have to show that the fibres of $U \to X$ are universally bounded. However, if Y is reasonable, then the fibres of $U \to Y$ are universally bounded, which immediately implies the same thing for the fibres of $U \to X$. Hence the result for "reasonable". □

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